Figure 1: The Hydrogen drum machine

Debian-like systems), or by using a package manager.

```
sudo apt-get install hydrogen
```

Hydrogen's interface is simple – just like the apps you may have in your smartphone. Drum patterns are made by clicking on the screen. The sound library contains a large set of instruments including the Indian ones like the *tabla*. You can save the project for further editing, and export it as a MIDI or audio file.

Synthesisers, soundfonts and connection programs

Before you get into MIDI applications, make sure that your system has the following basic requirements: soundfonts, synthesisers and connectors. However, if you find this section difficult or boring, simply skip it, and start practical composing with MuseScore. Be sure to come back to this part when you feel you are ready.

A MIDI file has no pre-recorded sound in it (this is why it can't hold a vocal song). The file only contains some notations for different instruments used in that particular piece of music (and this is why it is so lightweight). The information in a MIDI file can be converted to sound using mathematical formulae. But the most realistic method is to use soundfonts or patches that have pre-recorded samples.

Distros like Ubuntu come with TiMidity++ as the default synthesiser. You can install another popular synthesiser called FluidSynth and its graphical front-end, Qsynth. Whenever you want your program to use FluidSynth to produce sound, start Qsynth and make appropriate connections using the facilities provided by your MIDI program or by using a connector like QJackCtl.

In order to get FluidSynth working, you need a soundfont like Fluid R3 GM. You can install the package *fluid-soundfont-gm* to get it on your computer. To configure Qsynth with this soundfont, start Qsynth, press the *Setup* button, and then open the soundfont file (*/usr/share/sounds/sf2/FluidR3_GM.sf2*) under the *Soundfonts* tab.

From the main window of QJackCtl, you can start the JACK server (if not started already), and take the connections window ('Connect'). The connection is made by selecting the

Figure 2: Qsynth

input port, the output port and pressing the *Connect* button.

VMPK: The virtual piano

Homes of the rich were once decorated with a grand piano. The digital counterpart to this massive instrument is a virtual piano on our smartphones! In both cases, most of the owners don't know how to use their keyboards properly. But having them is still a pleasure, and playing on them is fun.

Your search for a perfect virtual piano in GNU/Linux ends with VMPK (Virtual MIDI Piano Keyboard). The cross-platform virtual keyboard is not as silly as it seems. You can use it as a sophisticated MIDI controller, and even connect it to your physical music keyboard.

Simply run the following command to install VMPK on Debian-based systems. You may search for the package 'vmpk' using your preferred package manager on other distros.

```
sudo apt-get install vmpk
```

If it doesn't produce sound, you'll have to check the MIDI connections at *Edit->Connections*. From the *Output MIDI Connection* drop-down list, choose the preferred output. It can be a TiMidity port or a port provided by another synthesiser like FluidSynth. There will be no usable ports listed if you haven't started any synthesiser yet. Please refer to the 'Troubleshooting MIDI' section to learn how to do that.

MIDI connectivity

Composers who're just starting off always dream of connecting a physical keyboard to their PCs to create music. But it remains a dream for most people since the options are confusing and unlike software, experimenting with hardware can be an expensive game. Let's get the doubts cleared and start experimenting.

The first question is about the device. You can choose a MIDI controller or a synthesiser ('Musical keyboard') with MIDI support. MIDI controllers don't produce sound of their own, but provide a lot of pitch control/modulation facilities to instruct the software on your PC. The synthesiser can be used alone or as a controller for your software (if the device supports MIDI). But most models lack controlling features.

The second question is about the port. No normal computer has a built-in MIDI port. Then how does one connect his device with the PC? The answer is via the USB. Most MIDI controllers and synthesisers support a USB interface. It seems to be plug-and-play also. If there is no native USB support, you can get an adapter.